NBA Player Points: Pre-Game Distributional Forecast

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A pre-game model for player points totals built on possession-level data. Produces a Negative Binomial distribution from which over/under probabilities at any line can be read directly. Reasonably calibrated on average across lines; benchmarked against multiple model families.

1 Task and Framing

Given 5.49M rows of possession-level NBA data, the goal is to predict each player's total points in a game before tip-off, with a distributional forecast so over/under probabilities at any line are immediate. The final model is

$$\hat{f}(y \mid \mathbf{x}) \;=\; \mathrm{NB}\Big(\hat{\mu}(\mathbf{x}),\, \hat{\sigma}^2(\mathbf{x})\Big)\,,$$

where $\hat{\mu}$ is V3, a LightGBM regressor with Tweedie loss ($p = 1.5$), and $\hat{\sigma}^2 = \hat{\mu} + \alpha(\hat{\mu})\hat{\mu}^2$ is an empirical mean-binned dispersion fit on a held-out validation fold. The choice fell out of a head-to-head comparison (Section 4) against LightGBM with alternative losses, XGBoost benchmarks, a hyperparameter-tuned variant, and naive baselines.

The dataset contains no DNP rows; a player appears only on possessions they were on the floor for. The model therefore estimates $\mathbb{E}[\text{points} \mid \text{player appears}]$, not $\mathbb{E}[\text{points} \mid \text{on roster}]$. In production this requires an upstream availability model fed by injury reports.

2 Data Reconnaissance

Rows (possession × player)	5,492,310
Possessions (exactly 10 rows each)	549,231
Games	2,787
Player-games (modelling grain)	**59,256**
Players	729
Teams	32
Date range	2022-10-18 to 2024-11-16
Target mean / median / max	10.3 / 8 / 70
Fraction of zero-point player-games	12.3%

Table 1: Verified data facts.

Time-played interpretation. `time_played` is cumulative seconds at the start of each possession a player is on the floor for (possession 1 always shows 0). So max(`time_played`)/60 slightly under-estimates true minutes by the duration of the player's last possession (∼0–25 s), which is small and non-systematic. Team totals confirm: $\sum_{\text{players}} \max($`time_played`$) \approx 14{,}430\,\text{s}$ per team per game (≈240 min, matching regulation).

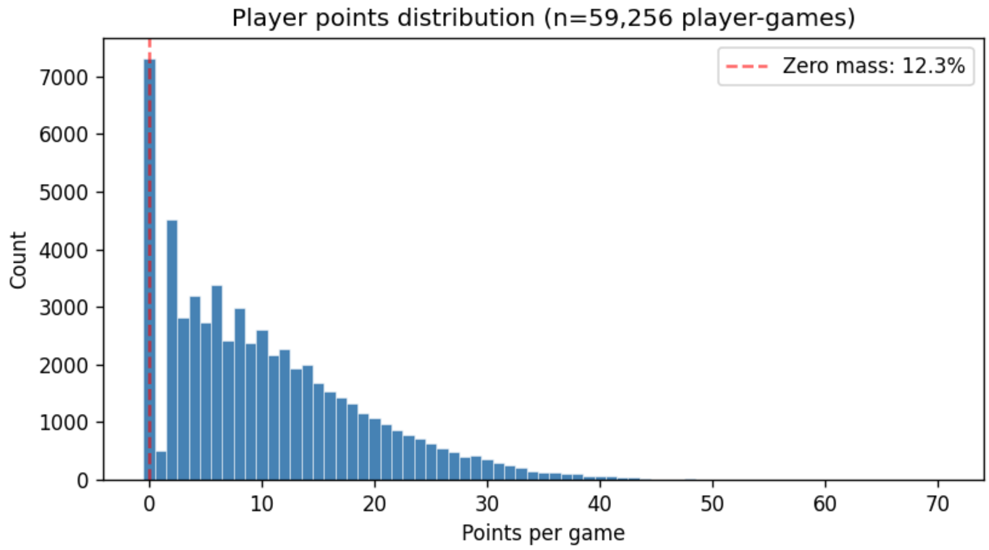

Figure 1: Player points per game. The spike at zero (12.3%) and the right skew with a max of 70 rule out Gaussian losses and motivate a count distribution.

3 Method

3.1 Aggregation

The possession-level table collapses to one row per (game, player). Box-score sums for points/rebounds/assists; max(`time_played`)/60 for minutes; n-unique on `possession_number` for on-floor exposure; modal position per game. Team-game pace and points-for/against are joined for opponent features.

3.2 Leakage discipline

For every feature value at row r to be admissible, it must be derivable from rows whose `game_date_time` is strictly less than r's, at the appropriate grain (player / team / opponent). Enforced by sort and `shift(1)`, grouped by (entity, season) so rolling windows reset across season boundaries. A unit test in `leakage_check.py` recomputes `pts_mean_season` and `pts_ewma_10` from scratch for random test rows; passes with zero errors.

3.3 Features (40, all pre-game)

Player form: EWMA of points at spans 5/10/20; season-to-date PPG; rolling 10-game std of points and minutes; EWMA-10 of minutes, FGA, FTA, 3PA; usage proxy $(\text{FGA} + 0.44 \cdot \text{FTA} + \text{TOV})/n_{\text{poss}}$; points-per-36 and points-per-minute; true-shooting% EWMA-20; team-attempt-share EWMA-10 (player FGA / team FGA); starts-proxy EWMA-10; games-played-so-far.

Context: home/away; days rest (clipped 0–14); back-to-back; three-in-four (third game in a four-day window, known pre-game from the schedule); games in last 5/14 days; position (7-way and 3-way G/F/C buckets).

Opponent and team (rolling 20, season-scoped): opponent rolling def-rating, off-rating, pace, FGA/3PA/FTA allowed per 100 possessions, OT-rate; opponent rolling points allowed to player's 3-way position bucket; player's team rolling pace and off-rating; expected pace (harmonic mean of team and opponent pace).

Schedule: `team_game_idx_in_season` (at unique team-game grain so every player on a team shares the same index), the opponent counterpart, and `month`.

Identity features (tested, not kept): `team_id`, `opp_id`, and `player_id` as raw categoricals were evaluated; see Section 4 for results.

3.4 Loss choice and dispersion

Tweedie loss ($p = 1.5$) is a compound Poisson-Gamma whose support matches the target (zero mass and positive continuous). It marginally beats Poisson and RMSE in the comparison below. Dispersion is fit on the validation fold by binning the predicted-mean decile and computing per-bin residual variance, then solving $\alpha_{\text{bin}} = (\text{var} - \mu)/\mu^2$ for the NB2 shape parameter (clipped to ≥ 0; under-dispersed bins fall back to Poisson). At inference, nearest-μ-bin lookup gives $\sigma^2 = \mu + \alpha\mu^2$.

4 Head-to-head comparison

Same temporal test set (last 20% of player-games by date, $n = 11{,}368$) for every variant. CRPS uses a closed-form discrete sum truncated at $K = 80$; the CDF comes from a Negative Binomial with the empirical mean-binned dispersion fit on the validation fold (identical distributional plumbing for all variants).

Variant	MAE	RMSE	CRPS
V0 Global mean baseline	6.974	8.651	—
V1 10-game EWMA PPG (strong baseline)	4.511	5.984	—
V2 EWMA(minutes) × EWMA(pts/min)	4.525	6.016	—
V3 LightGBM Tweedie (numeric only)	4.380	**5.760**	**3.068**
V4 LightGBM Poisson	4.396	5.764	3.070
V5 LightGBM RMSE	4.419	5.779	3.082
V6 LightGBM Tweedie + raw categoricals	4.385	5.762	3.070
V7 LightGBM Tweedie + `player_id`	4.432	5.809	3.099
V8 LightGBM Tweedie (Optuna-tuned, 25 trials)	**4.355**	5.797	3.108
V9 XGBoost Tweedie	4.382	5.807	3.099
V10 XGBoost Poisson	4.402	5.761	3.070

Table 2: All variants on the same test set ($n = 11{,}368$). V3 is best by CRPS, V8 is best by MAE. XGBoost and LightGBM land in the same neighbourhood; the predictive ceiling is feature-bound, not framework-bound.

4.1 What the comparison says

Loss-function axis (V3–V5). Tweedie ≈ Poisson > RMSE by small margins. Loss choice is third-order; features dominate.

Raw categoricals (V6). Adding raw `team_id`, `opp_id`, `position_code` is a no-op; the engineered rolling stats already absorb the team and position signal.

Player identity (V7). Adding raw `player_id` hurts (MAE 4.380 → 4.432, CRPS 3.068 → 3.099). The model memorises training-set players and fails to generalise across season boundaries where roles shift.

Optuna tuning (V8). A 25-trial TPE study over an 8-dim search space (Tweedie power, learning rate, leaves, min-data, subsampling, bagging frequency, L2) optimising validation MAE produces the best test MAE (4.355), a 0.6% improvement over V3. CRPS is worse (3.108 vs. 3.068). The tuned configuration is aggressive (low `min_data_in_leaf`=54, large `num_leaves`=201), producing a sharper conditional mean at the cost of variance calibration. Pick V8 if MAE is the objective, V3 if CRPS is.

XGBoost (V9, V10). Land in essentially the same place as the corresponding LightGBM variants, confirming the result is not a framework-specific artefact.

4.2 Two-stage minutes × scoring (V2)

A naive EWMA(minutes) × EWMA(pts/min) baseline gives MAE 4.525, worse than plain EWMA-PPG (4.511). Multiplying two noisy estimates compounds error. A proper joint model would help most under regime shifts (a star resting, a backup starting), which require injury and lineup data the dataset does not have. Minutes-related features (`min_ewma_10`, `min_std_10`, `starts_proxy_ewma_10`) are fed to the trees so the model can learn the interaction directly.

5 Distributional layer

5.1 Plain NB (final)

The variance is $\sigma^2 = \mu + \alpha(\mu)\mu^2$, with α fit empirically on the validation fold's residual variance per μ-bin. The Negative Binomial CDF gives closed-form $P(Y > L)$ for any line L.

5.2 ZINB (tested, not adopted)

A binary LightGBM predicts $\hat{p}_0 = P(Y = 0 \mid \mathbf{x})$. This is converted into a zero-inflation gate:

$$\text{gate} \;=\; \text{clip}\left(\frac{\hat{p}_0 - \text{NB}(0; \mu, \sigma^2)}{1 - \text{NB}(0; \mu, \sigma^2)},\ 0,\ 1 \right),$$

so the mixture's marginal $P(Y = 0)$ equals $\hat{p}_0$ by construction.

Layer (on the V3 mean)	CRPS	PIT KS-dev
Plain NB (FINAL)	**3.068**	**0.041**
ZINB	3.074	0.064

Table 3: ZINB shifts mass to 0 without preserving marginal mean, which shrinks $P(Y > L)$ by factor $(1 - \text{gate})$ and over-corrects at most lines. The PIT improvement at 0 does not offset the loss elsewhere. Plain NB is kept.

A hurdle model (separate NB re-fit on $Y > 0$) is the natural next attempt at zero-inflation; it is in the future-work list.

6 Final results

6.1 Headline metrics

V3 with plain NB dispersion gives MAE 4.380, RMSE 5.760, and CRPS 3.068 on the held-out test set of 11,368 player-games.

6.2 Over/under at sportsbook lines

Average predicted probability tracks the empirical rate within ~1 pp at five of six lines; the worst gap is at line 9.5 (−2.3 pp, mid-range scorers slightly under-predicted to go over).

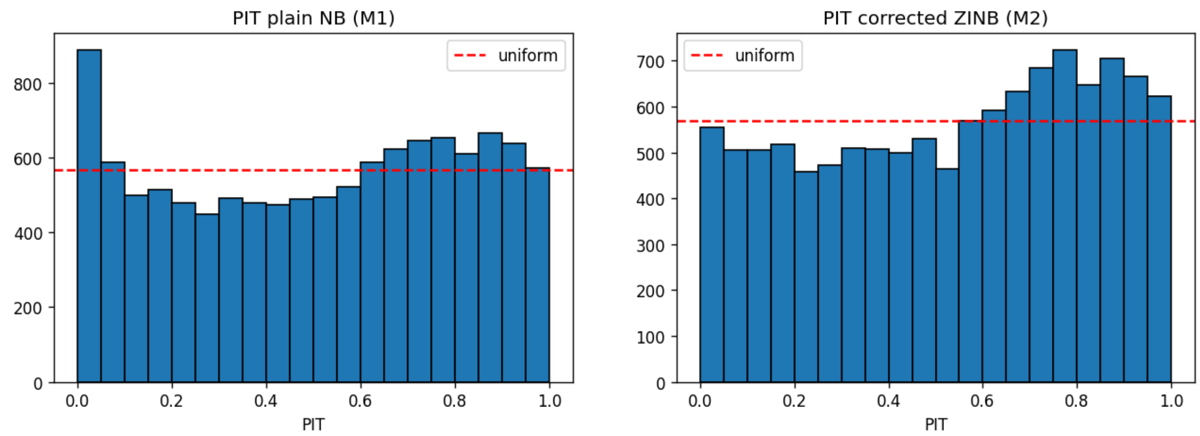

Figure 2: Randomised PIT histograms. Plain NB (left) has a residual spike at 0. ZINB (right) eliminates the spike but introduces broader humps in the 0.6–0.95 range. Net CRPS is negative.

Model	MAE	RMSE	CRPS
EWMA-10 baseline	4.511	5.984	—
V3 LightGBM Tweedie + plain NB [FINAL]	**4.380**	**5.760**	**3.068**
V8 LightGBM Tweedie (Optuna-tuned, best MAE)	4.355	5.797	3.108

Table 4: Test set, $n = 11{,}368$. V3 beats the EWMA baseline by 2.9% MAE and 3.7% RMSE. V8 wins on MAE but loses on CRPS.

Line	Empirical $P(Y > L)$	Avg predicted	Brier	Log-loss
4.5	0.677	0.673	0.142	0.436
9.5	0.449	0.426	0.149	0.457
14.5	0.264	0.253	0.116	0.368
19.5	0.145	0.140	0.079	0.257
24.5	0.072	0.071	0.048	0.161
29.5	0.032	0.032	0.026	0.091

Table 5: Over/under performance at six sportsbook lines.

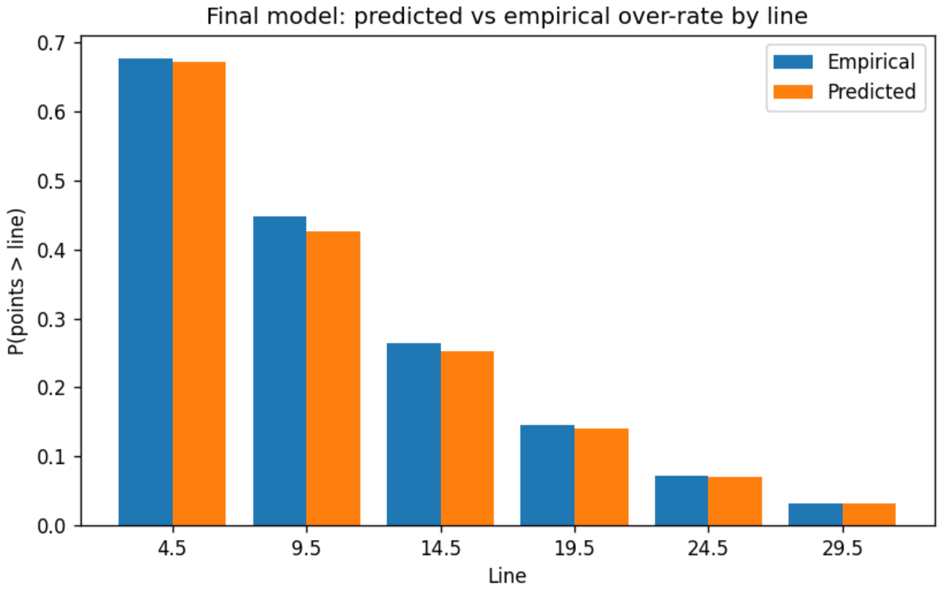

Figure 3: Predicted vs. empirical over-rate by line. Mild under-prediction at line 9.5; otherwise close to parity.

6.3 Reliability and PIT

Calibration is reasonable on average across lines, with mild under-confidence in the 0.3–0.6 predicted-probability range at lines 9.5 and 14.5 (empirical exceeds predicted by 2–5 percentage points there).

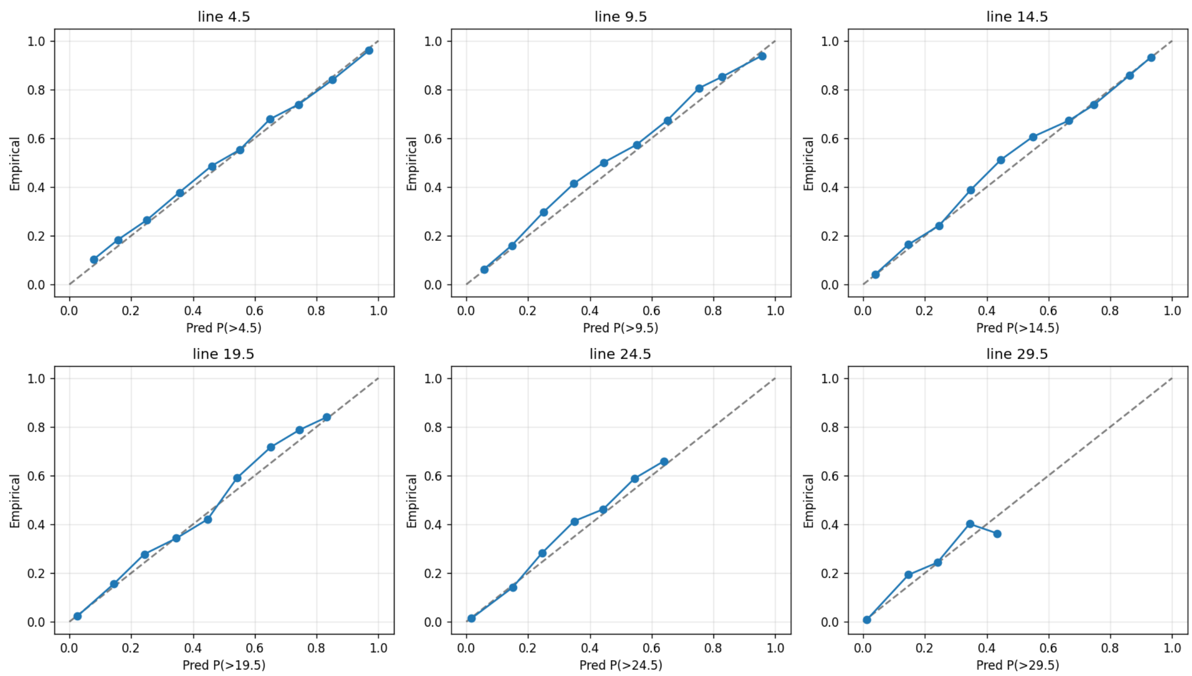

Figure 4: Reliability diagrams at all six lines.

6.4 Segment-level point errors

Errors break down cleanly by predicted-mean bucket. Predicted means match empirical means within ∼0.5 points in every bucket, and MAE grows roughly linearly with μ, consistent with the

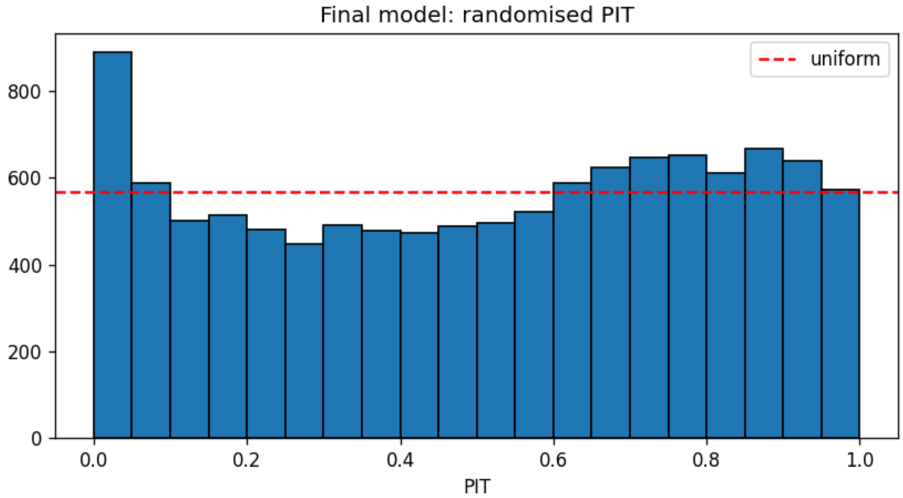

Figure 5: Randomised PIT for the final model. The distribution is broadly uniform with a residual spike at 0, consistent with the NB slightly under-modelling the zero mass.

heteroscedastic structure of count data.

Predicted-μ bucket	n	MAE	RMSE	emp. mean	pred. mean
≤ 5	3,007	2.83	3.72	3.26	3.24
5–10	3,709	4.20	5.34	7.63	7.37
10–15	2,302	4.94	6.23	12.33	12.25
15–20	1,271	5.67	7.17	17.42	17.43
> 20	1,079	6.59	8.34	23.91	23.41

Table 6: Errors by predicted-mean bucket.

7 Feature importance

Volume features dominate the LightGBM gain ranking, confirming that points = volume × efficiency and volume is much more stable game-to-game than efficiency. Schedule and rest features contribute meaningfully; opponent rolling features rank below volume because team-aggregate defence is a coarse proxy.

8 Final recommendation

V3 LightGBM Tweedie with numeric features and plain NB dispersion is the recommended configuration. The head-to-head in Section 4 establishes this concretely: tuning gives a 0.6% MAE gain at the cost of worse CRPS; XGBoost lands in the same neighbourhood; adding raw player identity memorises rather than generalises; ZINB underperforms plain NB on the distributional score that matters for pricing. Player points are inherently noisy at the game level, and the comparison shows that the irreducible floor cannot be moved by architectural complexity. The gains come from careful feature engineering with leakage discipline, not from model sophistication.

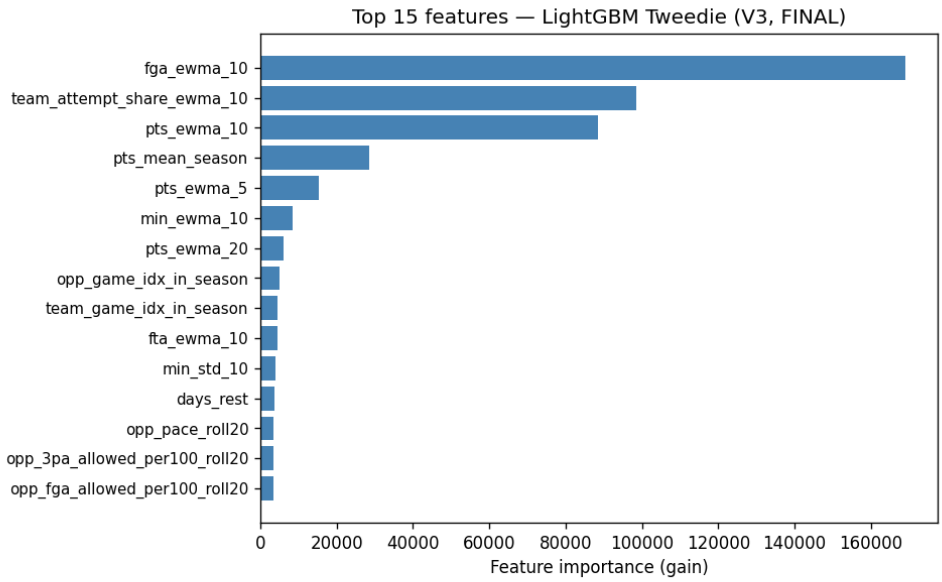

Figure 6: Top 15 features of V3 by gain. FGA EWMA, team-attempt-share, and minutes EWMA lead; schedule features and rest follow.

9 Limitations

No DNP rows (model is conditional on appearance); no injury, lineup, or market data; opponent defence aggregated to a 3-way position bucket only; mild calibration miss at line 9.5 (-2.3 pp); residual PIT spike at 0 (ZINB attempt did not pan out; hurdle model is the next attempt); player roles shift across seasons faster than EWMAs can adapt; pace approximated from possession counts rather than explicit game minutes.

10 What I would do with more time

(1) **Hurdle model.** Separate NB re-fit on $Y > 0$, the unattempted version of zero-inflation. Most likely to close the PIT zero-spike without the mean-preservation problem of ZINB.

(2) **Temperature scaling.** A single Platt scalar trained across all lines to correct the -2.3 pp bias at line 9.5.

(3) **Hierarchical shrinkage.** Shrink low-sample-size players toward position and role priors to reduce variance on the test-season tail.

(4) **MCMC-based hierarchical model.** Replace per-player EWMAs with a full Bayesian posterior over player skill using MCMC (e.g. NUTS via NumPyro). This pools information across players with different sample sizes and produces principled uncertainty estimates.

(5) **Quantile regression ensemble or conformal prediction intervals.** Non-parametric distributional alternatives.

(6) **SHAP analysis.** Verify that high-importance features behave sensibly across the input range and surface unexpected interactions.

(7) **Possession-sequence player embeddings.** A small transformer over recent possessions to learn skill vectors that aggregate rate stats cannot represent.

Reproducibility

```
python3 src/01_aggregate.py        # -> artifacts/data/player_game.parquet
python3 src/02_features.py         # -> artifacts/data/features.parquet
python3 src/leakage_check.py       # unit test, 0 errors
python3 src/03a_compare_models.py  # -> artifacts/models/ artifacts/eval/
python3 src/03b_tune.py            # -> artifacts/models/optuna_best_params.json
python3 src/03c_retrain_v8.py      # -> artifacts/models/ artifacts/eval/
python3 src/03d_xgboost.py         # -> artifacts/models/ artifacts/eval/
python3 src/04_zinb_calibrate.py   # -> artifacts/data/ models/ eval/
python3 src/05_final_eval.py       # -> artifacts/eval/ artifacts/plots/
```

Each script is commented with the rationale for every decision. Intermediate parquet artefacts are persisted so individual steps can be re-run independently.